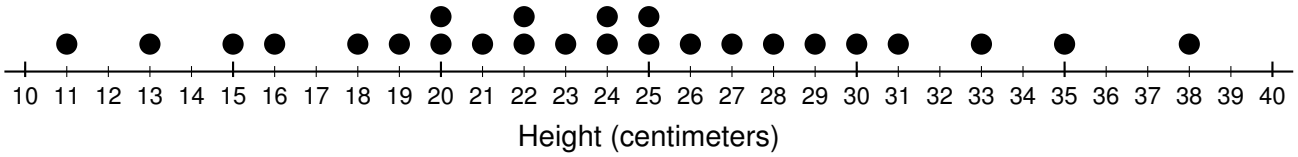
Unit 1 Test: Displaying One Variable KEY

Name _____

Show your work.

1. **I make a histogram.**

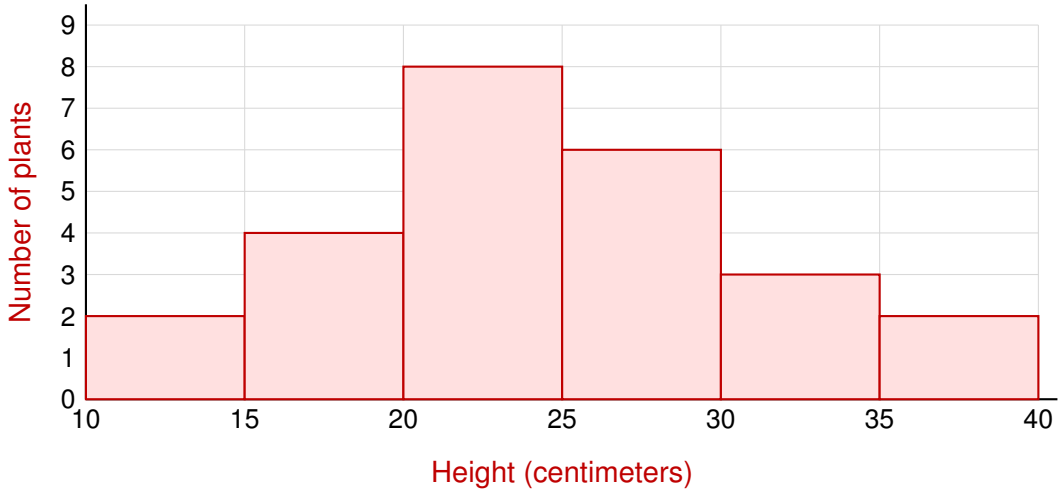
   A class grew 25 bean plants. The dot plot shows the height of each plant.



   (a) Make a frequency table. Count how many plants are in each group. Remember the histogram rule.

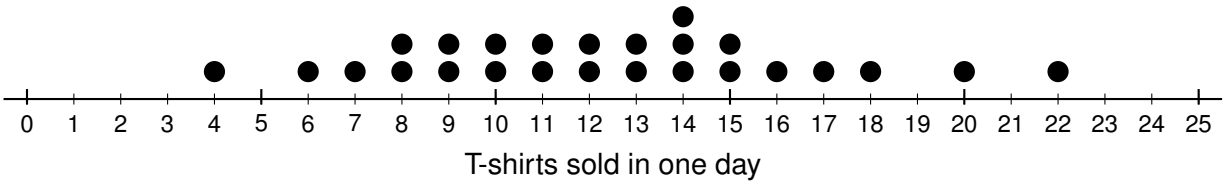
Height	Frequency
10 to 15	2 (11, 13)
15 to 20	4 (15, 16, 18, 19)
20 to 25	8 (20, 20, 21, 22, 22, 23, 24, 24)
25 to 30	6 (25, 25, 26, 27, 28, 29)
30 to 35	3 (30, 31, 33)
35 to 40	2 (35, 38)

   (b) Make a histogram. Label both axes.



2. **I find the five-number summary, and I make a box plot.**

A store kept track of T-shirt sales for 25 days. The dot plot shows how many T-shirts it sold each day.

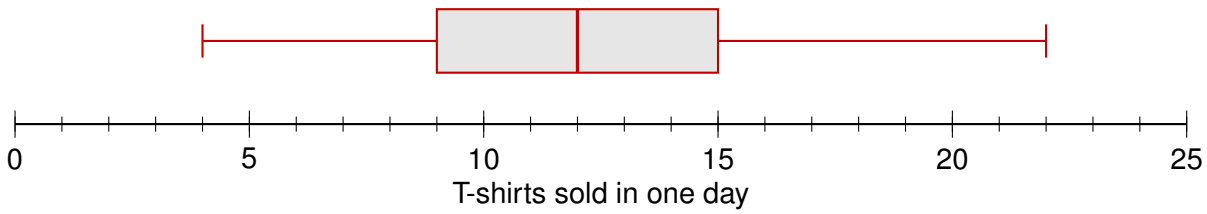


(a) Find the five-number summary.

n = 25. The median is the 13th value: 12. Leave it out.
Lower half: the first 12 values. Q1 = (9 + 9) ÷ 2 = 9 (6th and 7th values)
Upper half: the last 12 values. Q3 = (15 + 15) ÷ 2 = 15 (19th and 20th values)

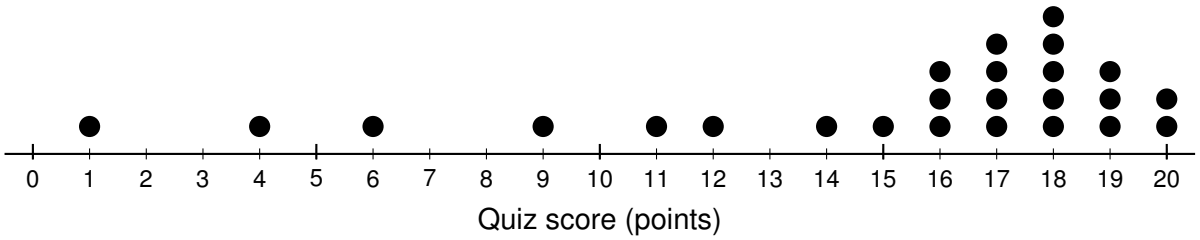
Min 4 Q1 9 Median 12 Q3 15 Max 22

(b) Make a box plot.



3. **I find the mean and the median, I tell the shape, and I choose the best measure of center.**

25 students took a quiz. The quiz had 20 points. The dot plot shows each student's score.



(a) Find Σx and the mean, $\bar{x}$. (Σ means add up. n is the number of values.)

$\Sigma x = 1 + 4 + 6 + 9 + 11 + 12 + 14 + 15 + 3(16) + 4(17) + 5(18) + 3(19) + 2(20) = 375$

$n = 25, \quad \bar{x} = \dfrac{\Sigma x}{n} = \dfrac{375}{25} = 15$ points

(b) Find the median.

$n = 25$. The median is the 13th value: 17 points.

(c) Circle the shape of the data.

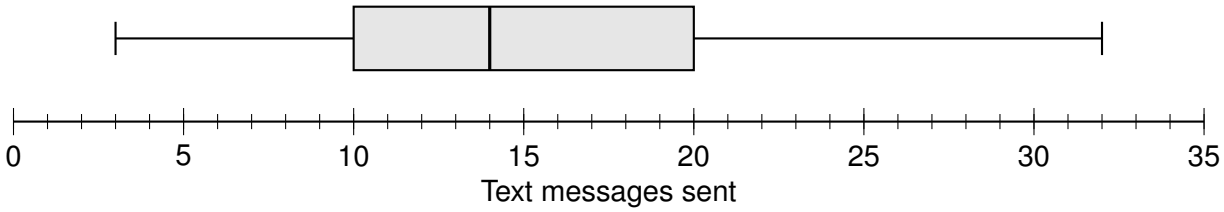
(skewed left) symmetric skewed right

(d) Which describes a typical score better, the mean or the median? Explain.

The median. The data are skewed left. A few low scores (1, 4, 6) pull the mean down to 15. 18 of the 25 students scored 15 or more, so the mean is too low to be typical.

4. **I use the 1.5 $\times$ IQR rule to find outliers.**

The box plot shows how many text messages 25 students sent yesterday.



One more student sent 60 text messages. Now there are 26 students.
Adding 60 does **not** change Q1, the median, or Q3.

(a) Find the IQR. Then find the lower fence and the upper fence.

From the box plot: Q1 = 10, Q3 = 20

IQR = $Q3 - Q1 = 20 - 10 = 10$

$1.5 \times 10 = 15$

Lower fence = $10 - 15 = -5$ Upper fence = $20 + 15 = 35$

(b) Is 60 an outlier? Explain.

Yes. 60 is above the upper fence, 35.

(c) How would the box plot change? What would stay the same? You do not have to draw a new box plot.

The box stays the same: Q1 = 10, median = 14, Q3 = 20.

The left whisker still ends at 3.

The right whisker still ends at 32. It stops at the largest value that is not an outlier (32 is below 35).

The only change: 60 is drawn as a dot by itself, far to the right.